

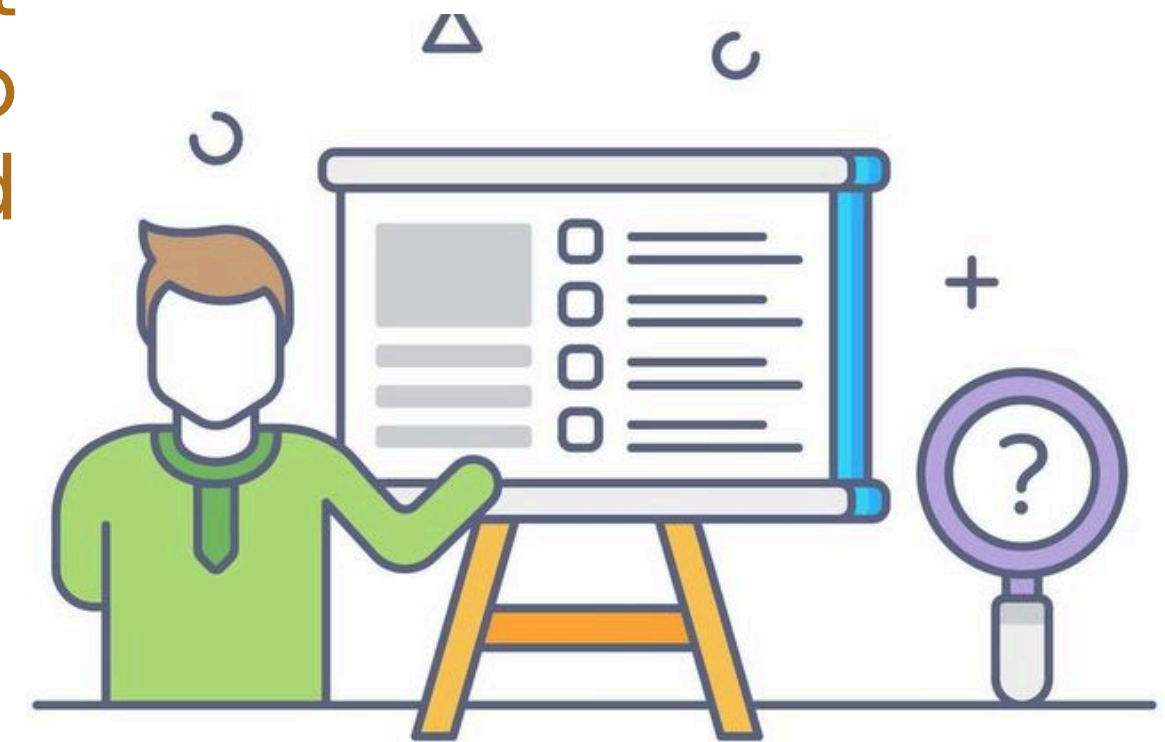
Project

Swiggy Data
Analysis Using SQL



Problem Statement

In a competitive food delivery ecosystem, data-driven decision-making is crucial for identifying top-performing restaurants and improving customer experience. This project uses advanced SQL techniques to analyze Swiggy restaurant data and extract meaningful business insights related to pricing strategy, competition, ratings, and delivery performance.



Why It Needs To Be Solved

- Identify top-performing restaurants.
- Improve delivery efficiency .
- Understand pricing strategy.
- Analyze city-wise competition.
- Support data-driven decisions.



Tools & Technology

MySQL – Data analysis and query execution

MySQL Workbench – Database management

Kaggle Dataset – Data source



SQL Techniques Applied

- ◆ Basic Query Operations

SELECT, WHERE, ORDER BY, LIMIT

→ Used for data extraction and filtering.

- ◆ Aggregation Functions

COUNT(), AVG(), MAX(), MIN(), SUM()

→ Used for city-wise statistical analysis.

- ◆ GROUP BY & HAVING

→ Performed grouped analysis and filtered aggregated results.

- ◆ Window Function

RANK(), DENSE_RANK(), ROW_NUMBER(), PARTITION BY

→ Ranked restaurants within each city to identify top performers.

- ◆ JOIN Operations

INNER JOIN, LEFT JOIN, RIGHT JOIN, SELF JOIN

→ Combined multiple tables for relational analysis.

- ◆ Subqueries

→ Applied advanced filtering and extracted top records.

- ◆ CASE Statements

→ Categorized restaurants based on price, ratings, and delivery performance.



Key Analytical Questions

1. What are the top 3 restaurants in each city based on order price ?
2. Which city has the highest average order pricing ?
3. Which restaurants fall under the premium price category ?
4. Expensive vs affordable Restaurant ?
5. Which restaurants provide the fastest delivery service ?
6. Which city has the highest number of restaurants ?
7. Top Restaurant by Ratings ?
8. Restaurant Categories by Ratings ?
9. Which restaurants have high ratings but relatively lower prices ?
(Value-for-money restaurants)
10. Are there restaurants that have high prices but low ratings ?
(Overpriced restaurants analysis)

Solving Questions Using SQL

Q.1 Top 3 restaurants in each city based on order price?

```
select Restaurant_Name, City, Order_Price,  
Rank() Over (partition by City order by Order_Price desc) as rnk  
from restaurent  
limit 3;
```

	Restaurent_ID	City	Restaurant_Name	Order_Price	rnk
▶	394904	Mumbai	Origami Japanese & Kore...	2500	1
	294499	Bangalore	Itc Windsor - Gourmet Co...	2500	1
	306434	Chennai	Malgudi - The Savera Hotel	2500	1

Q. 2 Which city has the highest average order pricing ?

```
select Restaurant_Name ,City, round(avg(Order_Price),2) as  
        average_price  
        from restaurent  
group by Restaurant_Name ,City , Order_Price  
order by Order_Price desc  
limit 5 ;
```

	Restaurant_Name	City	average_price
►	Origami Japanese & Kore...	Mumbai	2500.00
	Malgudi - The Savera Hotel	Chennai	2500.00
	Itc Windsor - Gourmet Co...	Bangalore	2500.00
	Yauatcha	Mumbai	2000.00
	Conrad Pune	Pune	2000.00

Q. 3 Which restaurants fall under the premium price category?

```
with premium_category as (  
  select Restaurant_Name , Order_Price  
    from restaurent  
  )  
  select *  
    from premium_category  
 where Order_Price > 1000;
```

	Restaurant_Name	Order_Price
►	Esplanade	1200
	Chef Inam'S Steak House	1200
	Taaareef	1100
	Tamarind	1100
	5 Spice	1100
	Prem'S	1200

Q. 4 Expensive vs affordable Restaurent

```
SELECT
a.Restaurant_Name as Expensive_Restaurant,
b.Restaurant_Name as afford_Restaurant,
a.City,
a.Order_Price as Expensive_Price,
b.Order_Price as afford_Price
FROM restaurent a
JOIN restaurent b
ON a.City = b.City
AND a.Order_Price > b.Order_Price
LIMIT 20;
```

	ensive_Restaurant	afford_Restaurant	City	Expensive_Price	afford_Price
▶	Bangalore Cafe	Tandoor Hut	Bangalore	400	300
	Villa Cafe	Tandoor Hut	Bangalore	1000	300
	oor'S Cafe	Tandoor Hut	Bangalore	500	300
	Ghar	Tandoor Hut	Bangalore	400	300
	na Burma	Tandoor Hut	Bangalore	1300	300
	Corner Restaurant	Tandoor Hut	Bangalore	450	300

Q. 5 Which restaurants provide the fastest delivery service?

```
SELECT
d.Restaurent_ID,
d.Delivery_time,
s.Restaurant_Name,
s.City
FROM swiggy as s
RIGHT JOIN delivery as d
on s.Restaurent_ID = d.Restaurent_ID
order by Delivery_time asc ;
```

	Restaurent_ID	Delivery_time	Restaurant_Name	City
▶	10273	20	Scoops	Hyderabad
	44058	21	Wah Ji Wah (Budhvihar)	Delhi
	79331	22	Sarvottam Restaurant	Surat
	108986	23	Nic Natural Ice Creams	Hyderabad
	214809	23	Hotel Niagara	Hyderabad
	4138	24	Grand Hotel	Hyderabad

Q. 6 Which city has the highest number of restaurants?

```
select City ,  
max(Restaurent_ID) as total_Restaurent  
from swiggy  
group by City  
order total_Restaurent desc ;
```

	City	total_Restaurent
►	Kolkata	1346

Q. 7 Top Restaurent by Ratings?

```
SELECT s.Restaurant_Name, s.City,  
       r.Avg_ratings  
FROM swiggy as s  
JOIN ratings as r  
on s.Restaurent_ID = r.Restaurent_ID  
ORDER BY r.Avg_ratings desc  
LIMIT 10;
```

	Restaurant_Name	City	Avg_ratings
►	Zorro - Milkshakes And Th...	Hyderabad	5
	Papacream	Pune	5
	The Asian Pavilion	Mumbai	5
	Keventers Ice Creamery	Kolkata	5
	Fat Tiger	Delhi	5
	Cafe Kokomo	Mumbai	5

Q.8 Restaurent Categories by Ratings

```
SELECT
  s.Restaurant_Name,
  r.Avg_ratings,
  CASE
    WHEN r. Avg_ratings >= 4.5 THEN 'Excellent'
    WHEN r. Avg_ratings >= 4.0 THEN 'Very Good'
    WHEN r. Avg_ratings >= 3.5 THEN 'Good'
    ELSE 'Average'
  END AS Rating_Category
FROM swiggy s
JOIN ratings r
ON s.Restaurent_ID = r.Restaurent_ID ;
```

Restaurant_Name	Avg_ratings	Rating_Category
Tandoor Hut	4.4	Very Good
Tunday Kababi	4.1	Very Good
Kim Lee	4.4	Very Good
New Punjabi Hotel	3.9	Good
Nh8	4	Very Good
Treat	4.5	Excellent

Q.9 Which restaurants have high ratings but relatively lower prices?

```
SELECT
s.Restaurant_Name,
r.Avg_ratings,
s.Order_Price
FROM swiggy s
JOIN ratings r
ON s. Restaurent_ID = r. Restaurent_ID
WHERE r.Avg_ratings >= 4.5
AND s.Order_Price <= 500
ORDER BY r.Avg_ratings DESC, s. Order_Price ASC;
```

	Restaurant_Name	Avg_ratings	Order_Price
►	Afresh	5	100
	Shree Krishna Juice Bar	5	150
	The Asian Pavilion	5	200
	Nivala Apka Apna	5	200
	Creams & Bites	5	200
	Tricky Shakes	5	250

Q. 10 Are there restaurants that have high prices but low ratings?

```
SELECT
s.Restaurant_Name,
r.Avg_ratings,
s.Order_Price
FROM swiggy s
JOIN ratings r
ON s.Restaurant_ID = r.Restaurant_ID
WHERE r.Avg_ratings <= 3
AND s.Order_Price >= 1000
ORDER BY r.Avg_ratings DESC, s.Order_Price ASC;
```

Restaurant_Name	Avg_ratings	Order_P
Golden Wave Restaurant	2.9	1000
Itc Hotels- Nutmeg The C...	2.9	1000
Quattro	2.9	1000
Coconut Grove	2.9	1000
Caravan Serai	2.9	1000

Solution of the Problem Statement

Through structured SQL analysis, the dataset was explored to derive meaningful business insights regarding restaurant performance, pricing trends, ratings, and delivery efficiency.

The analysis helped in :

1. • Identifying top-performing restaurants in each city
2. • Determining cities with the highest restaurant concentration
3. • Classifying restaurants into pricing categories (Budget, Mid-range, Premium)
4. • Evaluating the relationship between ratings and pricing
5. • Detecting value-for-money restaurants (High rating + Low price)
6. • Analyzing delivery efficiency across cities

Using advanced SQL techniques such as aggregation, window functions, joins, subqueries, and case statements, raw dataset was transformed into actionable business insights.

Conclusion

- The SQL-based analysis successfully transformed raw restaurant data into meaningful business insights.
- By applying aggregation, ranking, joins, and conditional logic, key patterns related to pricing, ratings, restaurant distribution, and delivery efficiency were identified.
- The project demonstrates strong command over SQL fundamentals and analytical thinking required for real-world data analysis tasks.



Conclusion